

433MHz Wireless RELAY SWITCH

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This cost-effective 433MHz wireless relay switch circuit allows remote relay control using an RF transmitter and receiver module. It is ideal for scenarios where wired connections are impractical or undesirable, such as remote lighting, wireless home automation, garage door control, or DIY IoT devices.

The system utilises a widely available, low-cost 433MHz RF transmitter-receiver pair. The transmitter sends digital on/off signals, which the receiver detects to trigger the relay.

The SPDT relay at the receiver end operates on a 5V supply, depending on the load and power source. The switch offers a typical range of 50-100 metres in open areas and can penetrate walls with reduced performance. It provides a practical and reliable solution for wireless control applications. Fig. 1 shows the author's receiver prototype.

Circuit and working

The circuit of this 433MHz wireless relay switch consists of two main sections. One is the transmitter, whose circuit diagram is shown in Fig. 2. The second is the receiver, whose circuit diagram is shown in Fig. 3. The transmitter section utilises a 433MHz RF transmitter module (TX1) with three pins: data, Vcc, and ground. The data pin is connected through a 1k resistor (R4) to a push-to-on switch (S1), which also connects to the +5V supply. The Vcc pin is directly powered

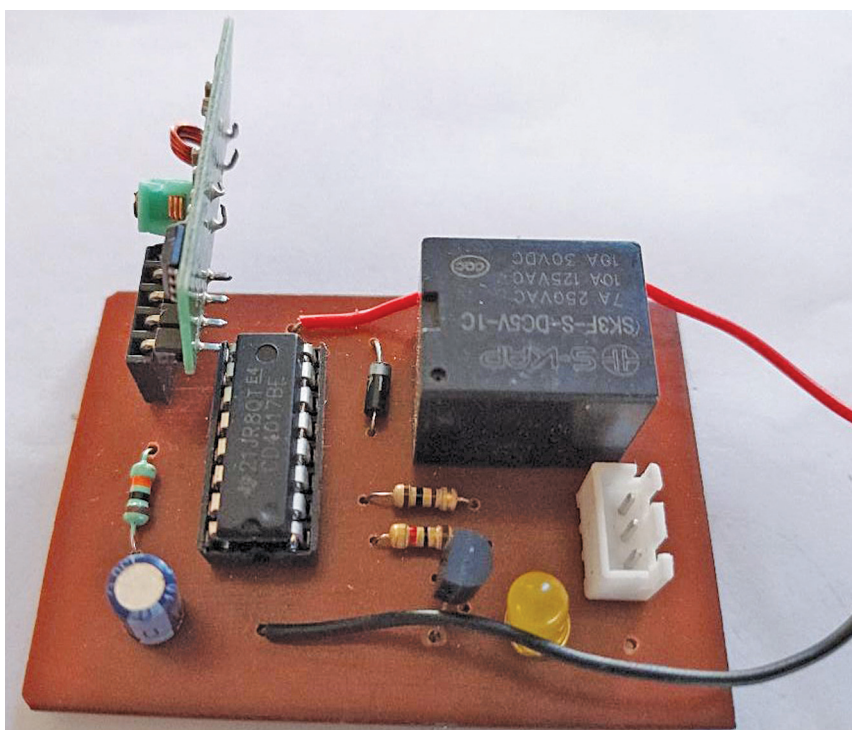


Fig. 1: Author's receiver prototype

by +5V, while the ground pin is connected to the system ground (power). To improve signal transmission, a 7cm wire can be connected to the antenna terminal of the transmitter module.

The receiver section includes a 433MHz RF receiver module (RX1) with four pins: Vcc, two data output pins, and ground. Both data output pins are connected to pin 14 (clock input) of the CD4017 decade counter IC, which is used to control the switching logic. Pin 14 is also connected to ground through a 56k resistor and a 1µF, 25V capacitor (C2) for signal conditioning. Pin 2 of the CD4017 (Q1

output) is connected to the base of an NPN transistor, BC547 (T1), via a 330-ohm resistor (R2). The transistor's emitter is connected to ground, and the collector is connected to one end of a 5V relay coil. A 5V supply also powers the other end of the relay coil. Flyback diode 1N4007 (D1) is placed across the relay coil with its anode connected to the transistor's collector and cathode to +5V to protect against voltage spikes.

The CD4017 is powered by +5V supply, with pins 8 and 13 grounded and pin 16 connected to Vcc. Pin 4 is connected to pin 15 (reset), enabling repeated toggling on each